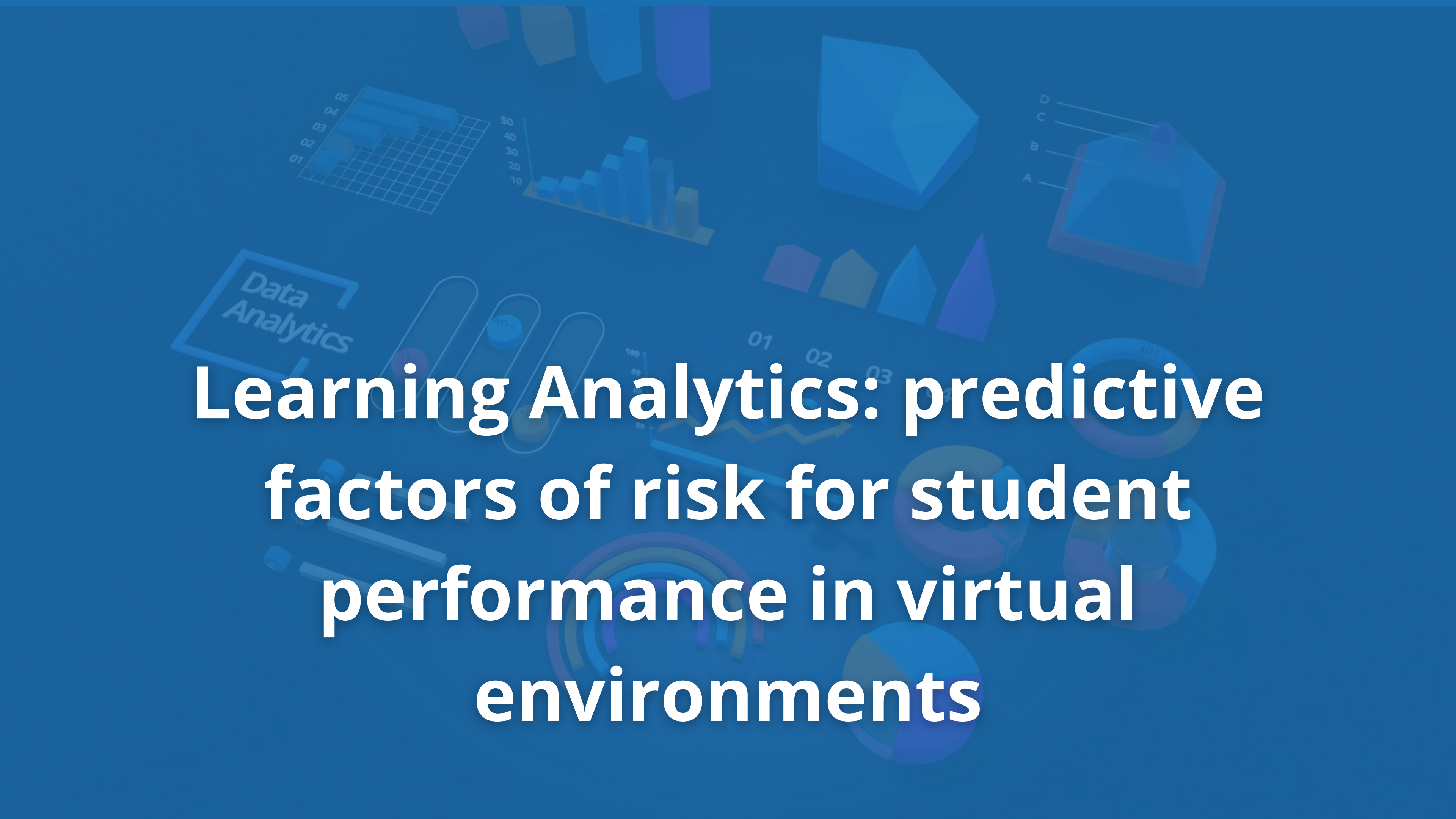




Learning Analytics: predictive factors of risk for student performance in virtual environments

Research Problem and Objectives



The fundamental objective of this study was to implement a Learning Analytics model capable of monitoring and predicting student performance, facilitating timely and personalized intervention.



The specific objectives included:

- Establishing the correlation between behavioral metrics in the Virtual Learning Environment (activity, persistence) and final academic performance.
- Developing a clustering system for the student cohort based on their interaction patterns and initial performance.
- Determining the critical thresholds of the identified variables that act as significant predictors of failure risk.



Context

The Open University is a British public university with the largest number of undergraduate students in the UK. It is the largest academic institution in the UK (and one of the largest in Europe), with 2 million students enrolled since its founding in 1969. As its name suggests, most of its students learn remotely.



Dataset

Open University Learning Analytics Dataset (OULAD). It contains data on courses, students, and their interactions with the Virtual Learning Environment (VLE) for seven selected courses (referred to as modules). The courses begin in February and October and are marked with the letters "B" and "J," respectively. The dataset consists of tables connected by unique identifiers.

Research Design



The study adopted a mixed-methods research design:

- Descriptive: Analysis of the distribution of performance and behavior variables.
- Predictive: Development of an analytical model to estimate the probability of success or failure based on early metrics.

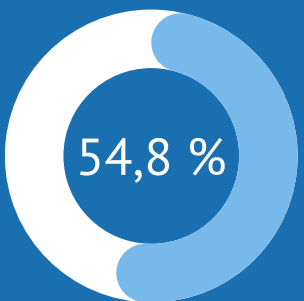


Analysis Tools and Techniques

Study Summary:

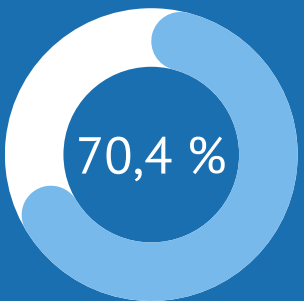
- Data pipeline and predictive modeling with Python (Pandas, Matplotlib, scikit-learn)
- Variable normalization and outlier handling
- K-Means for segmentation into C0–C3 by activity, persistence, and initial score
- Variable weights to distinguish C0 (success) from C1 (extreme risk)

Categorical variables



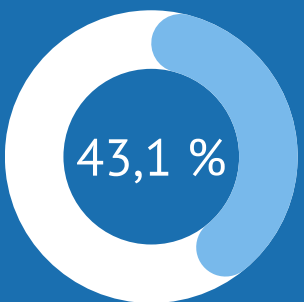
GENDER

Female



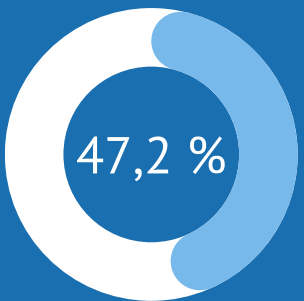
AGE

0-35 years



HIGHEST
EDUCATION

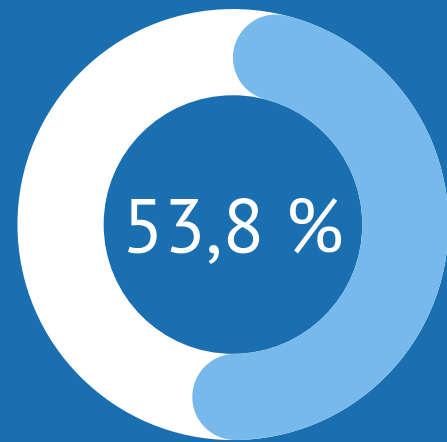
A level or equivalent



FINAL RESULT

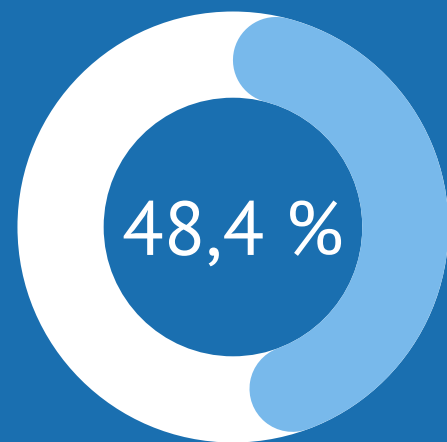
Pass / Distinction

Success / Risk by gender



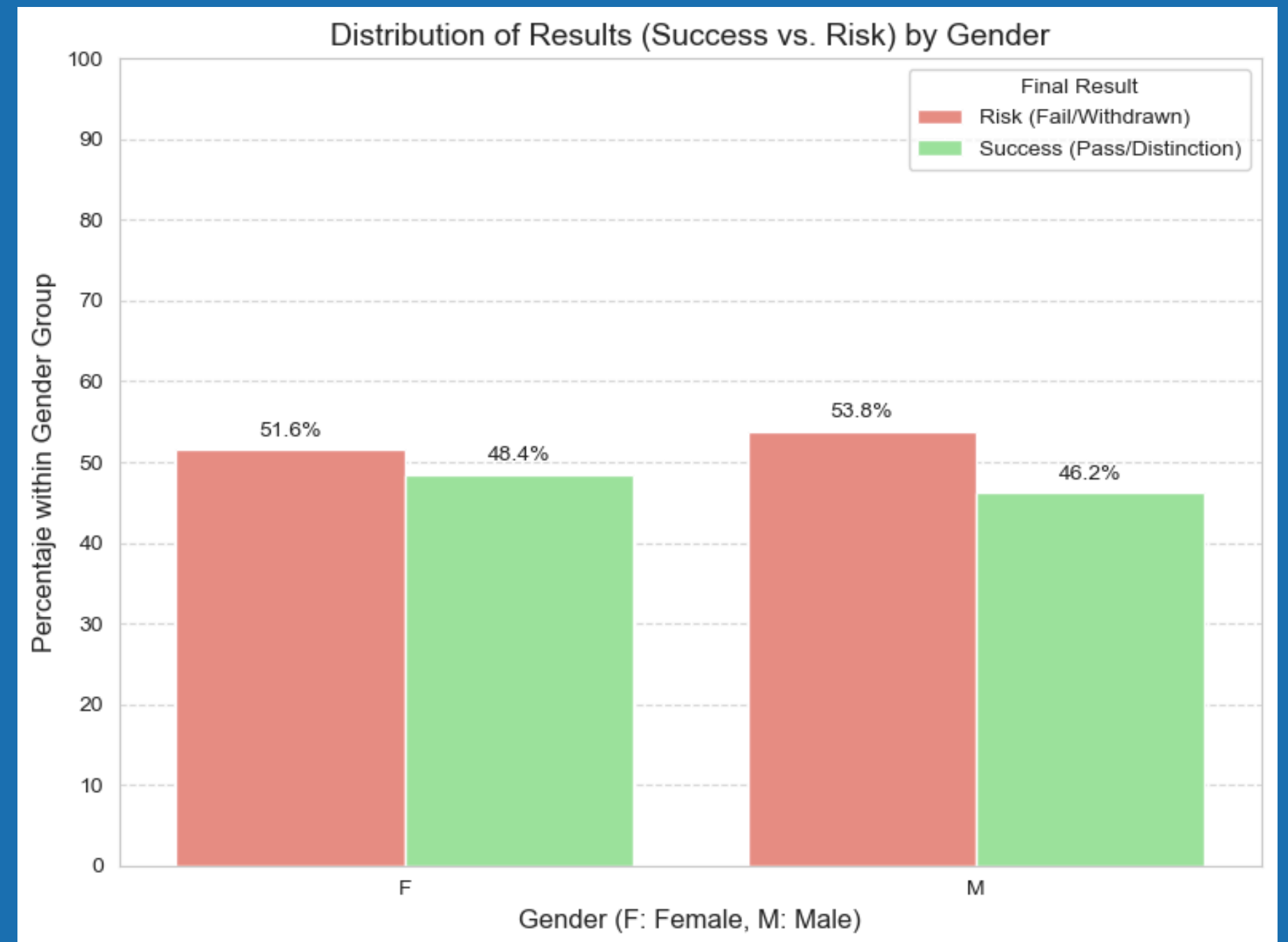
RISK GENDER

Mens are more predictable to fail

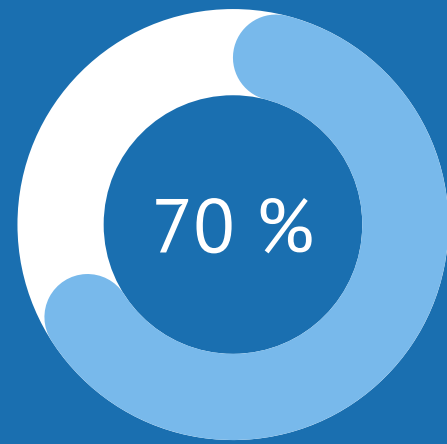


SUCCESS GENDER

Females have a little more changes to success

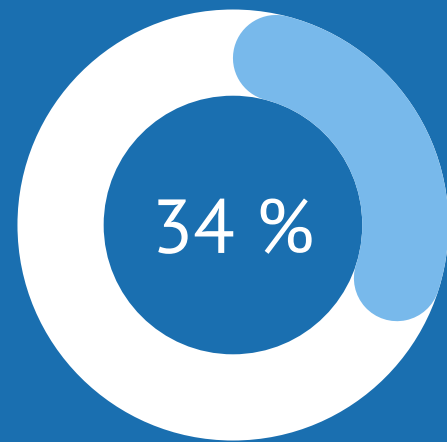


Previous education vs Risk rate



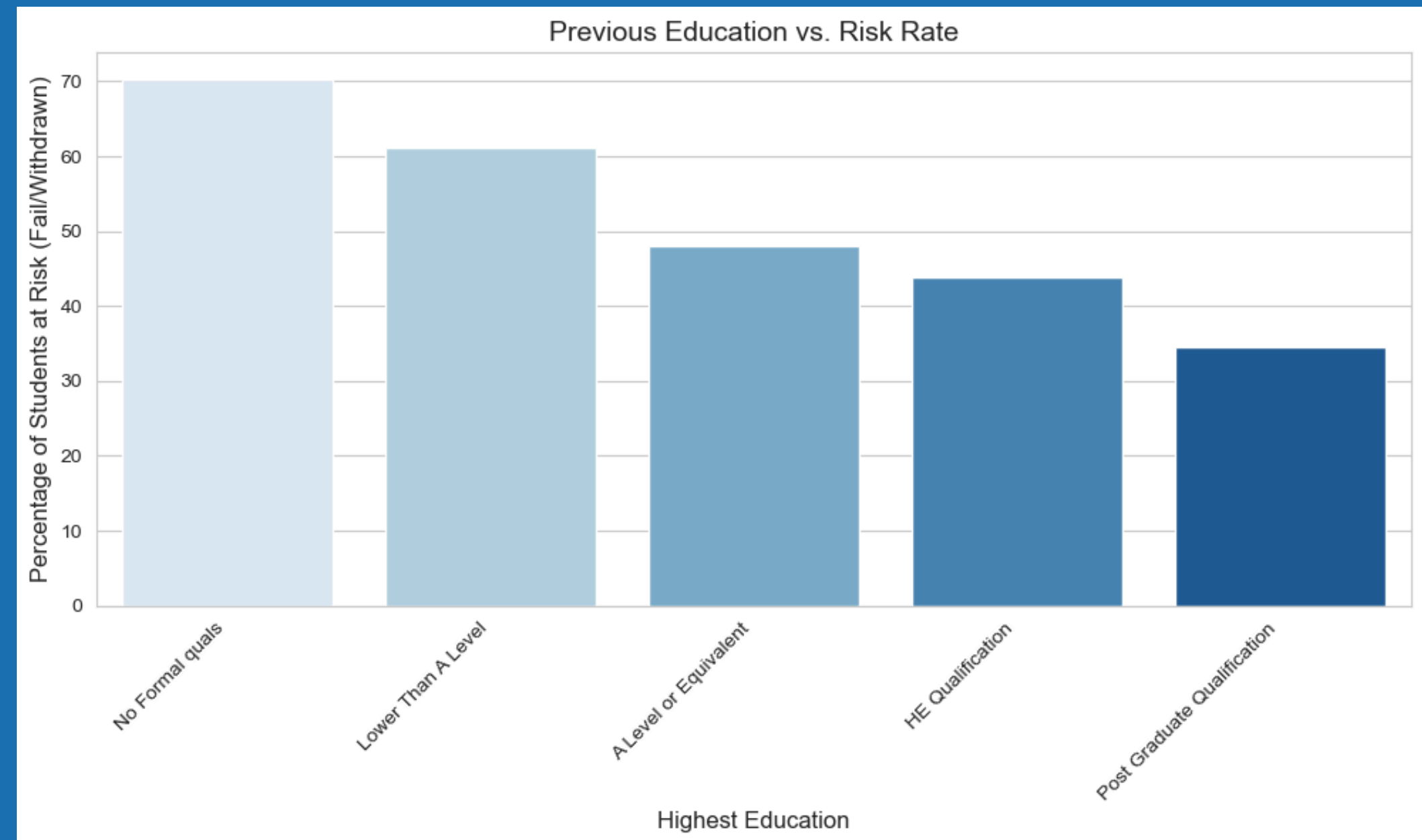
**NO FORMAL
QUALS**

refers to a non formal
education



POST GRADUATE QUAL

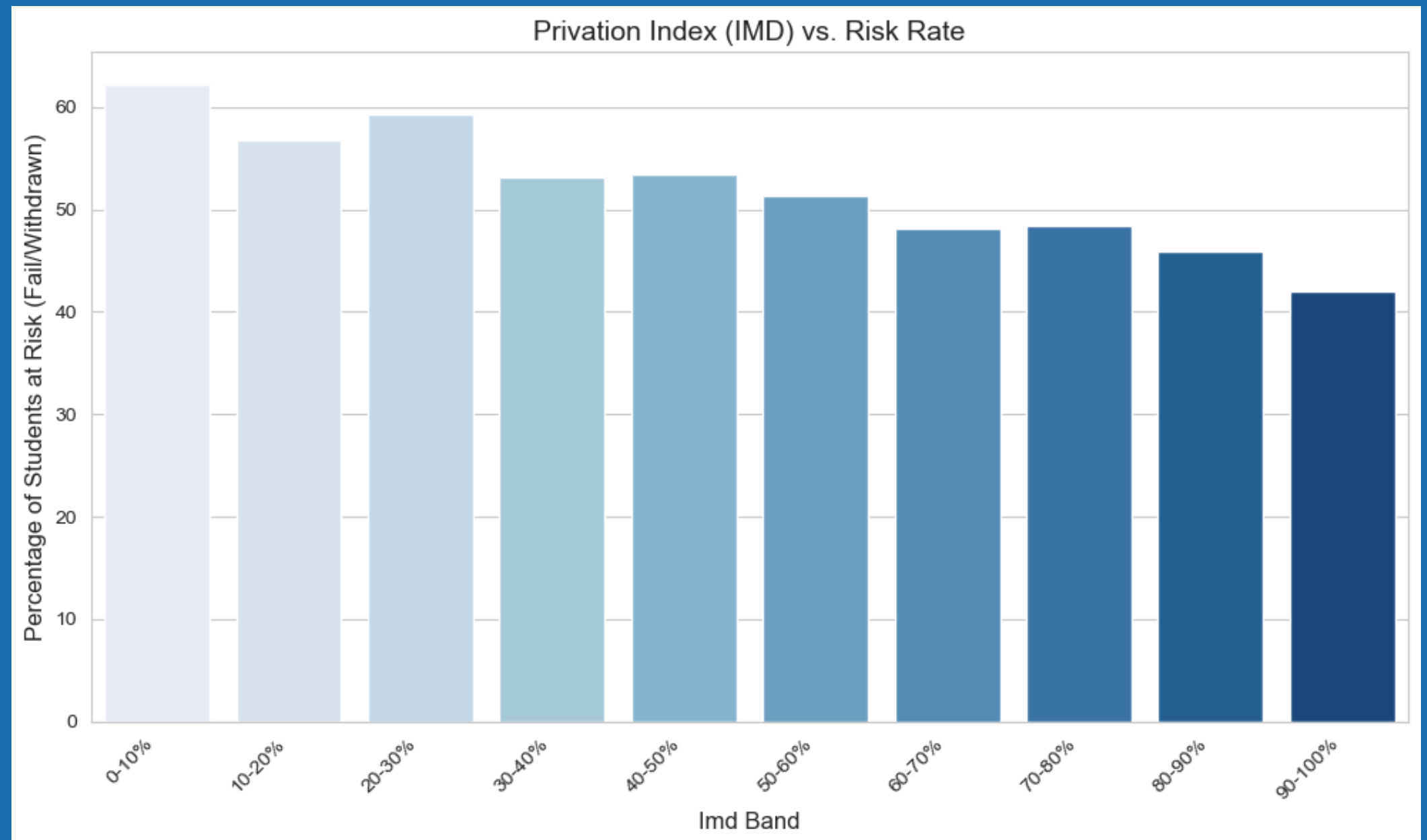
refers to a student who has
completed an undergraduate
degree



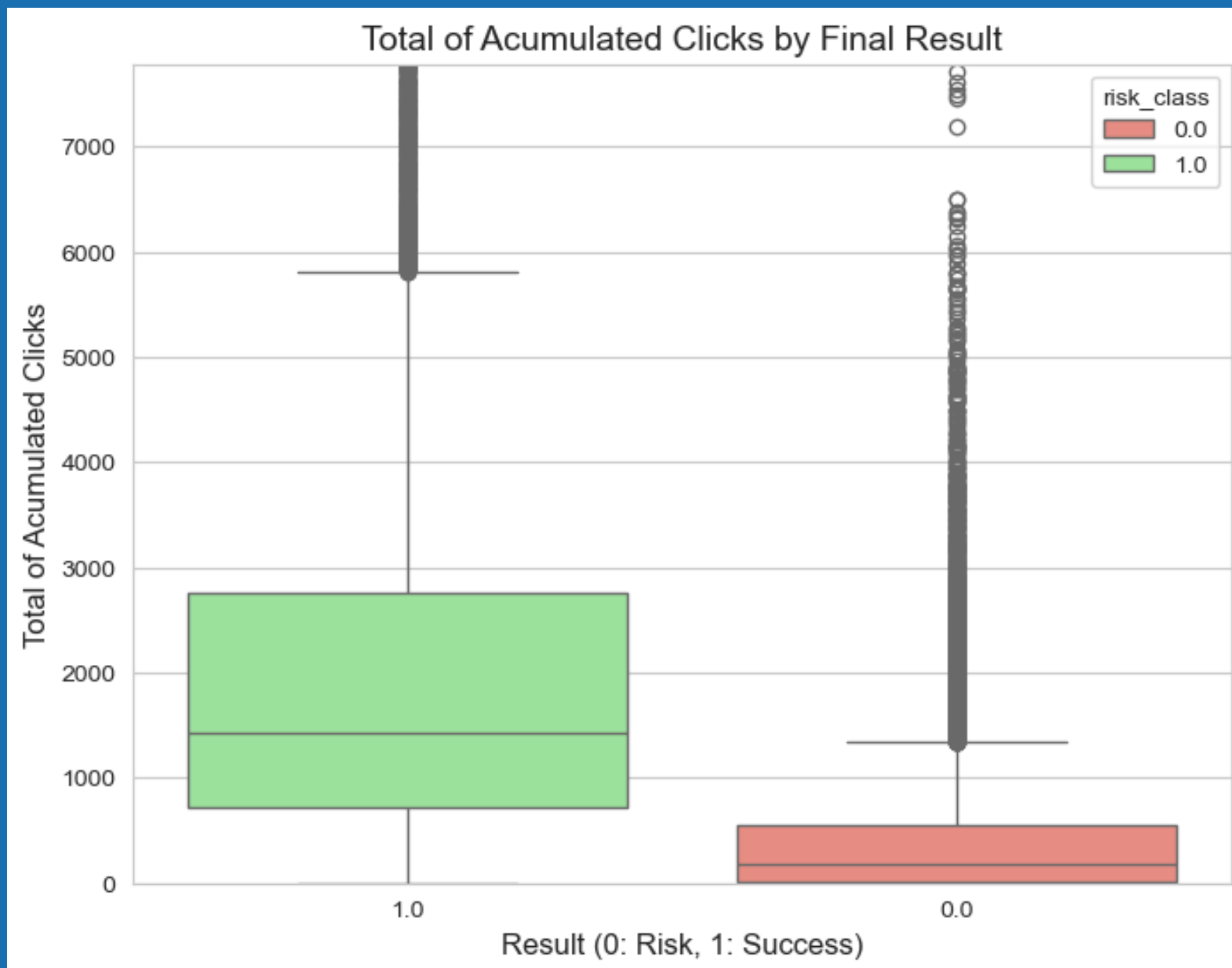
Privation index vs Risk rate

Index of Multiple Deprivation (IMD)

an official measure of relative deprivation for small areas in England, including income, employment, and health—to rank areas from most to least deprived, which helps authorities allocate resources.



Total clicks by final result



+ 1400

MAIN SIGNIFICANT DIFERENCE

Mean of clics much higher for successfull students



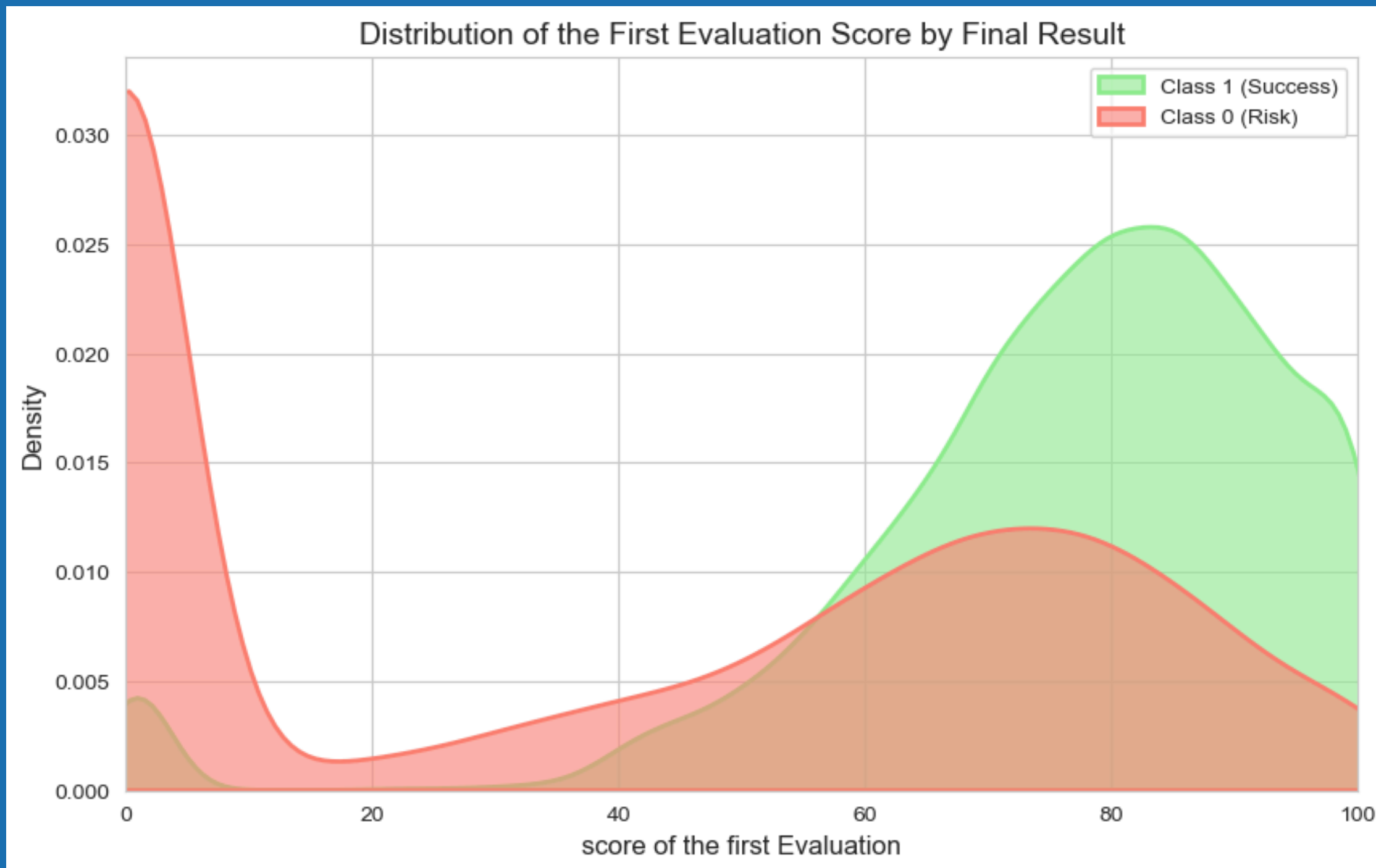
OUTLIERS

A small percentage of Risk users who have an unusually high click volume (even exceeding the median of the Success group)

INSIGTH

75% of Risk users have fewer clicks than the bottom 25% of Success users, indicating that click volume is a strong predictor of the final outcome.

Initial performance (Distribution of the first score)



0 - 20

HIGH PREDICTIVE VALUE FOR EXTREMES

Very low scores (approx. 0-20) are a strong indicator of risk

60 - 85

OVERLAP ZONE (AMBIGUITY)

The probability of success (green area) is greater than that of risk (red area), although the presence of at-risk students in this range suggests that the **first assessment** alone is not completely discriminatory.

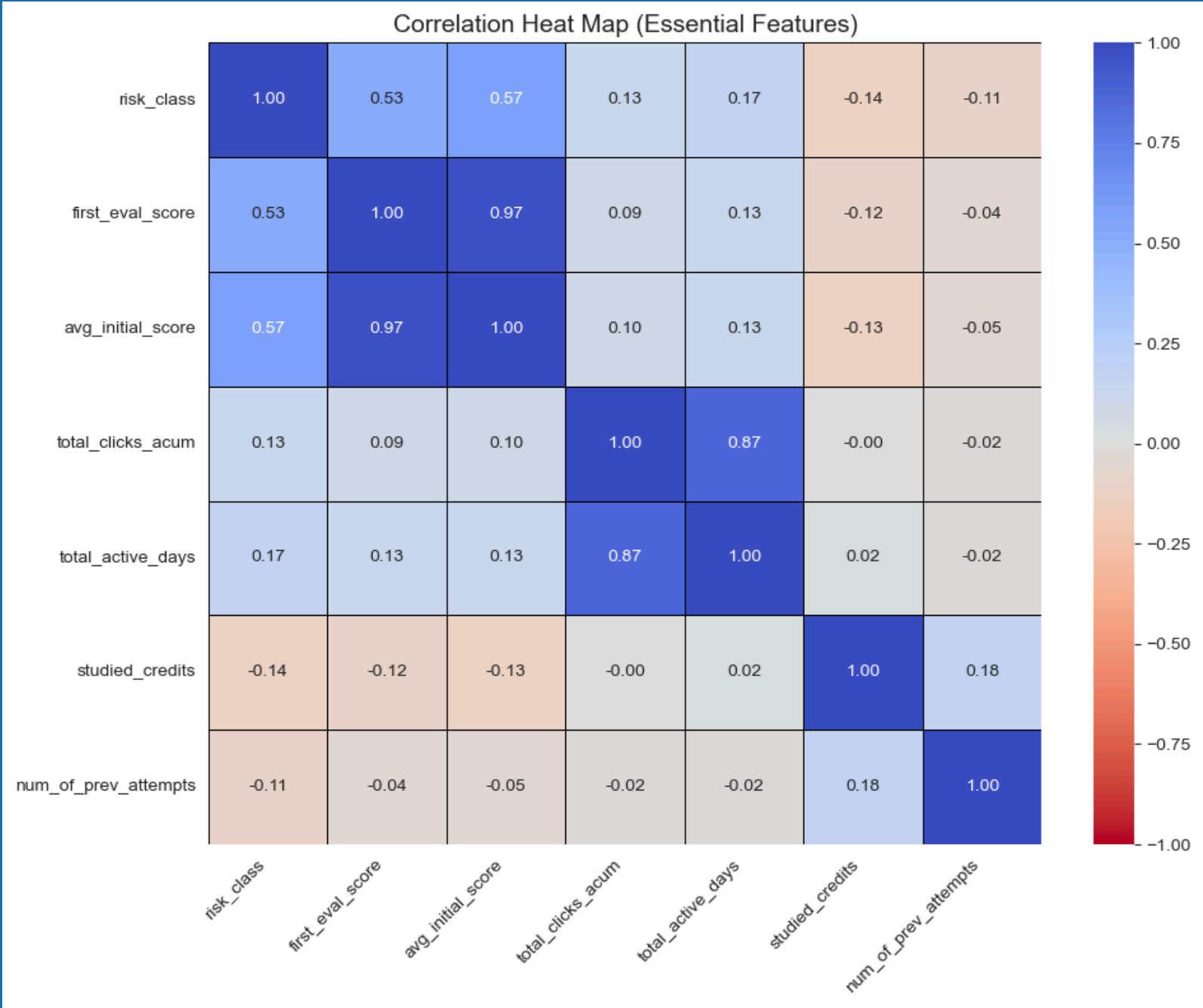
Correlation exploration: Heatmap



DATA ONE
total_active_days



DATA FOUR
total_clicks_acum



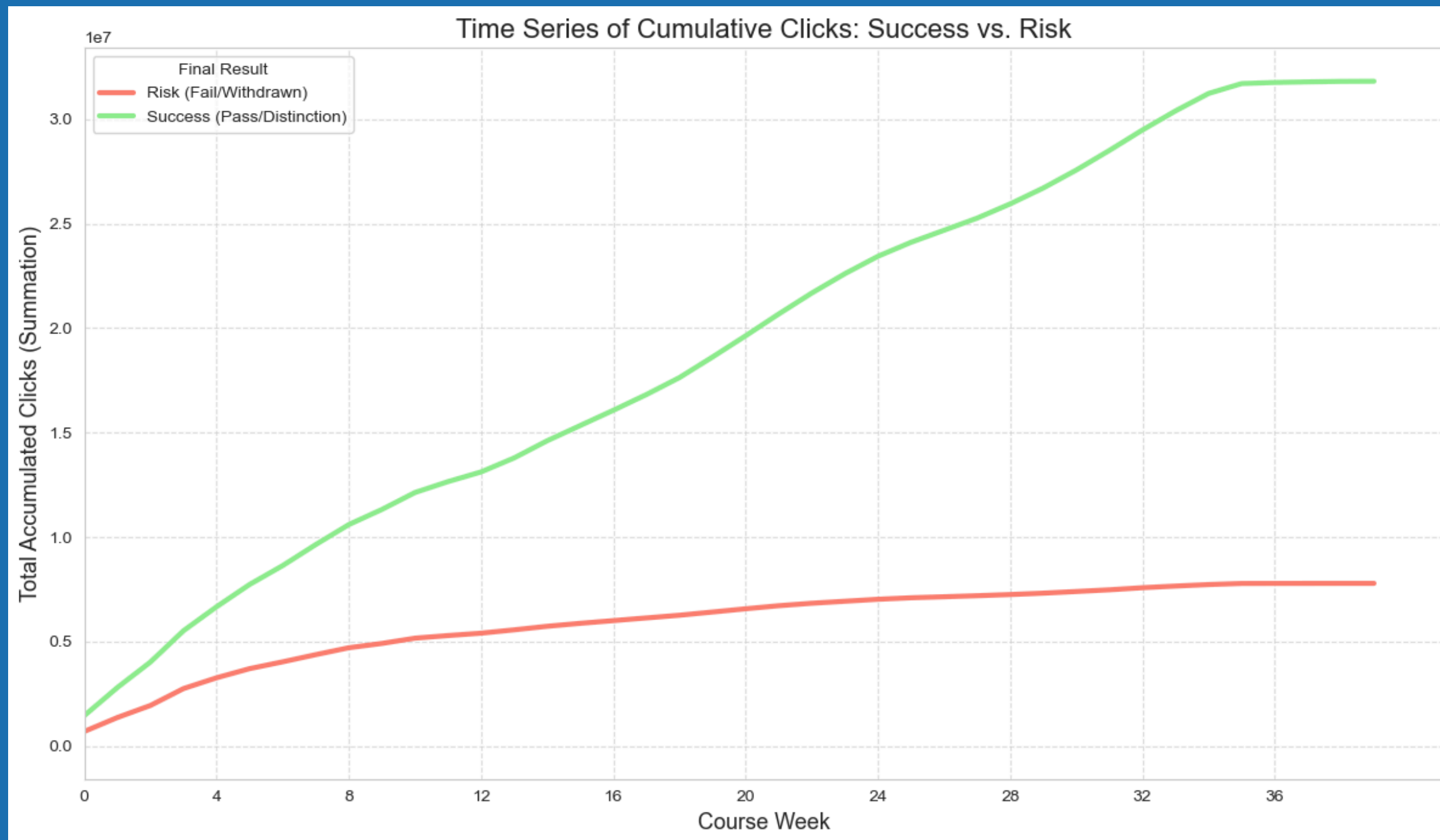
DATA TWO
avg_initial_score



DATA THREE
first_eval_score



Time Series of Cumulative Clicks per Result



0-4

SEPARATION PATTERN

The Risk curve diverge sharply from the Success curve. Express ideal point for early intervention.

32-36

FINAL ACTIVITY

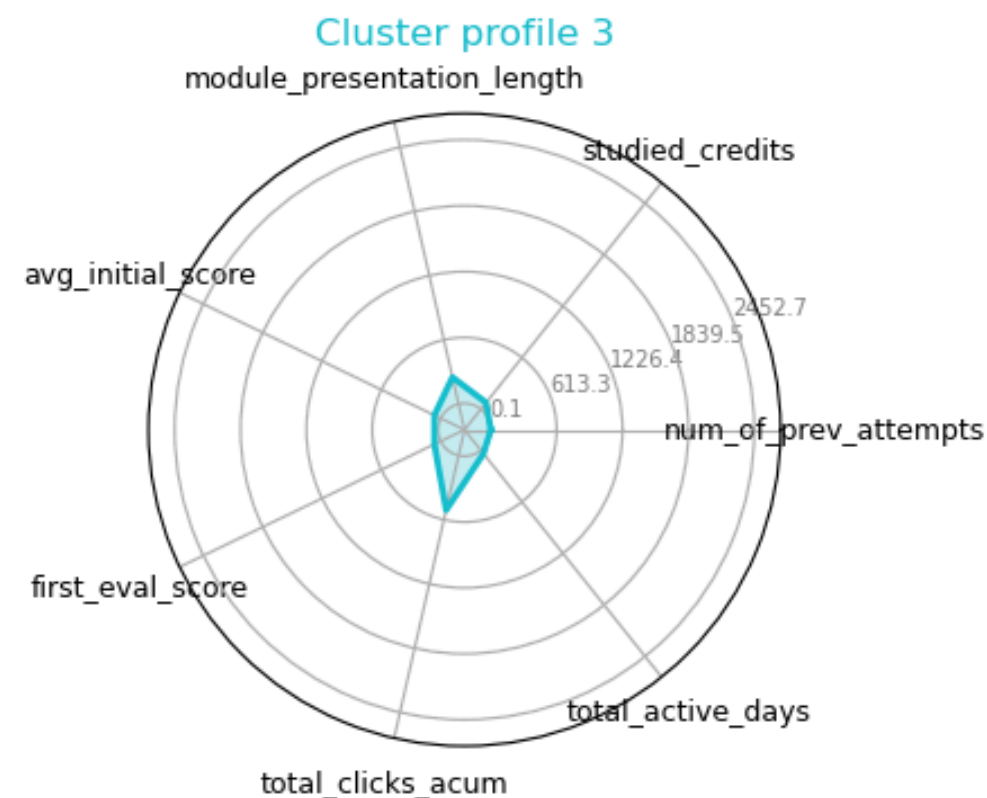
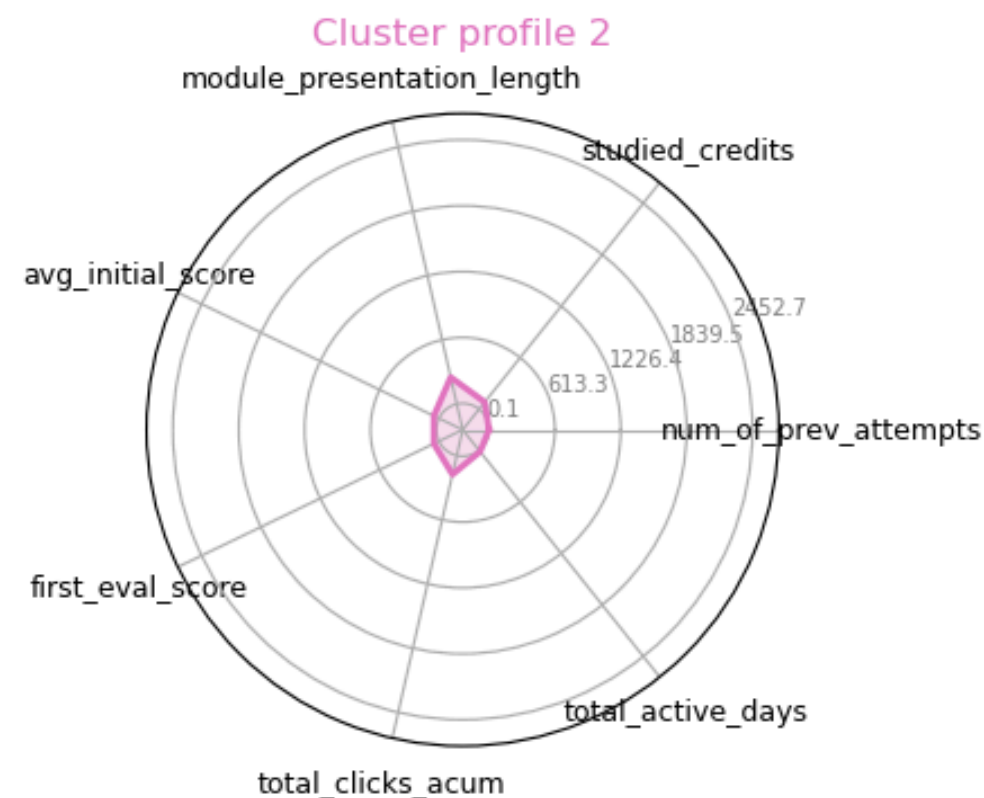
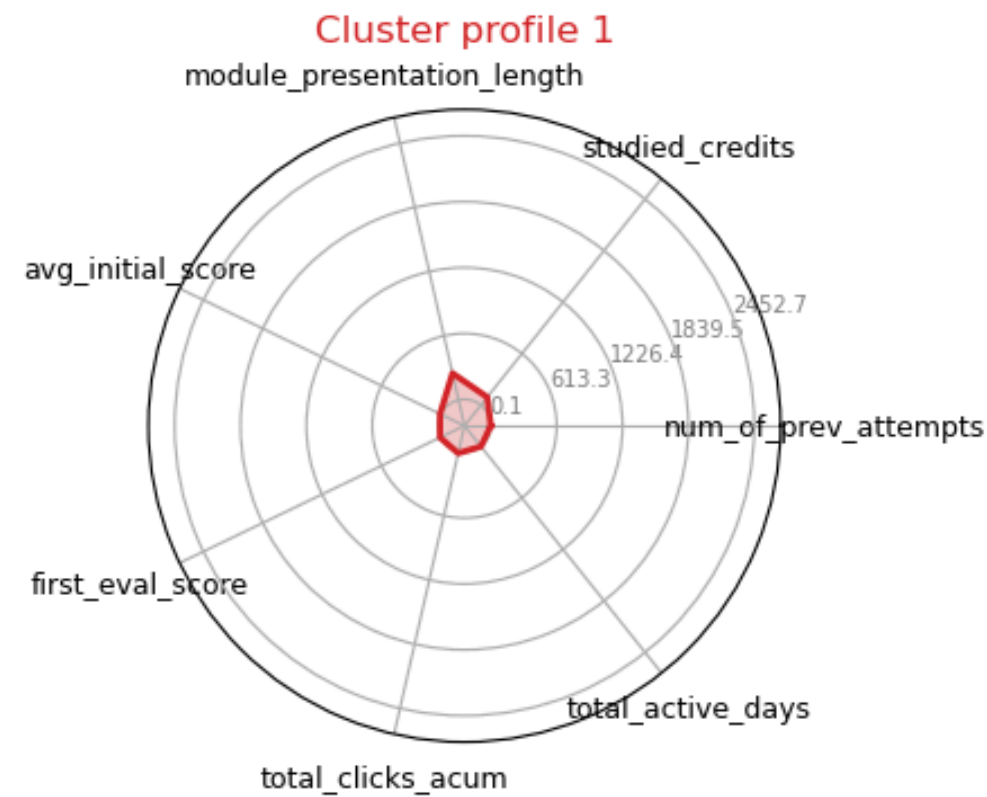
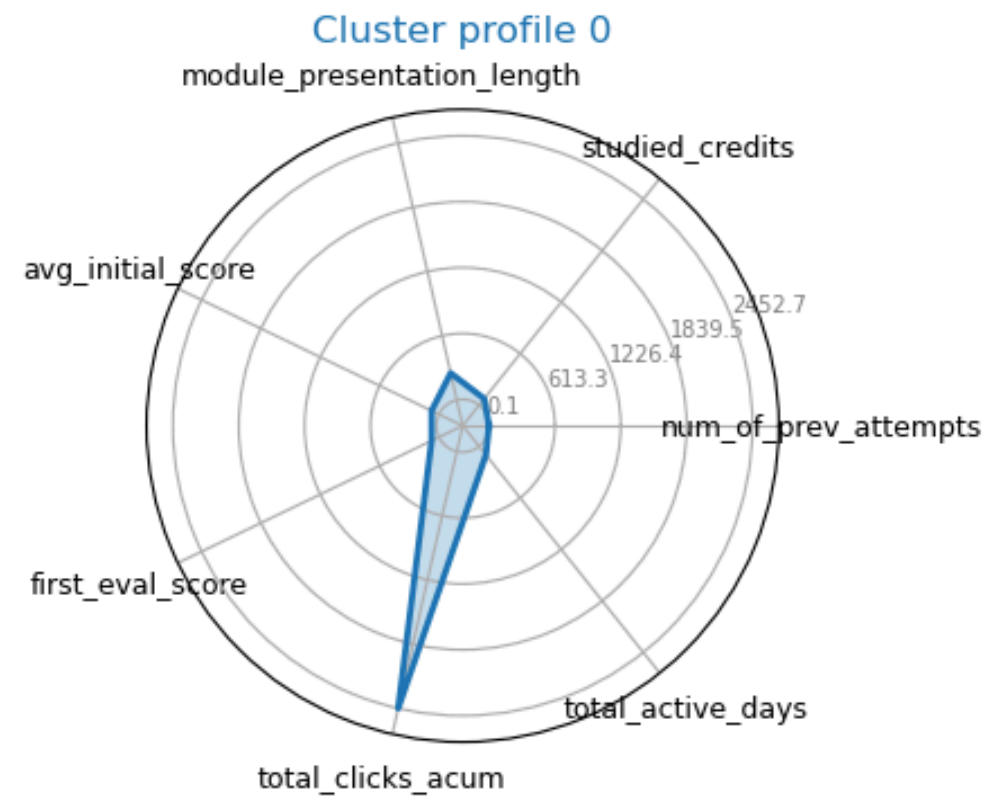
The gap at the end of the series is large, which is a key indicator that VLE activity is a very differentiating factor.

Classification model: Unsupervised Machine Learning

K-means grouping

The cluster profile (the centroids) indicates the distinctive characteristics of each group. We use radar charts as tools to visualize these profiles.

Students profile based on 4 Clusters



Application of K-Means and Cluster Assignment

CLUSTER 0 (SUCCESS)

High initial and first evaluation scores, many clicks, and active days. Low risk.

High-achieving and very active students.

CLUSTER 2 (HIGH RISK)

Low scores, few clicks, and active days. **Students with low commitment and insufficient results. Very high risk.**



CLUSTER 1 (EXTREME RISK)

Very low initial and first evaluation scores, few clicks, and active days.

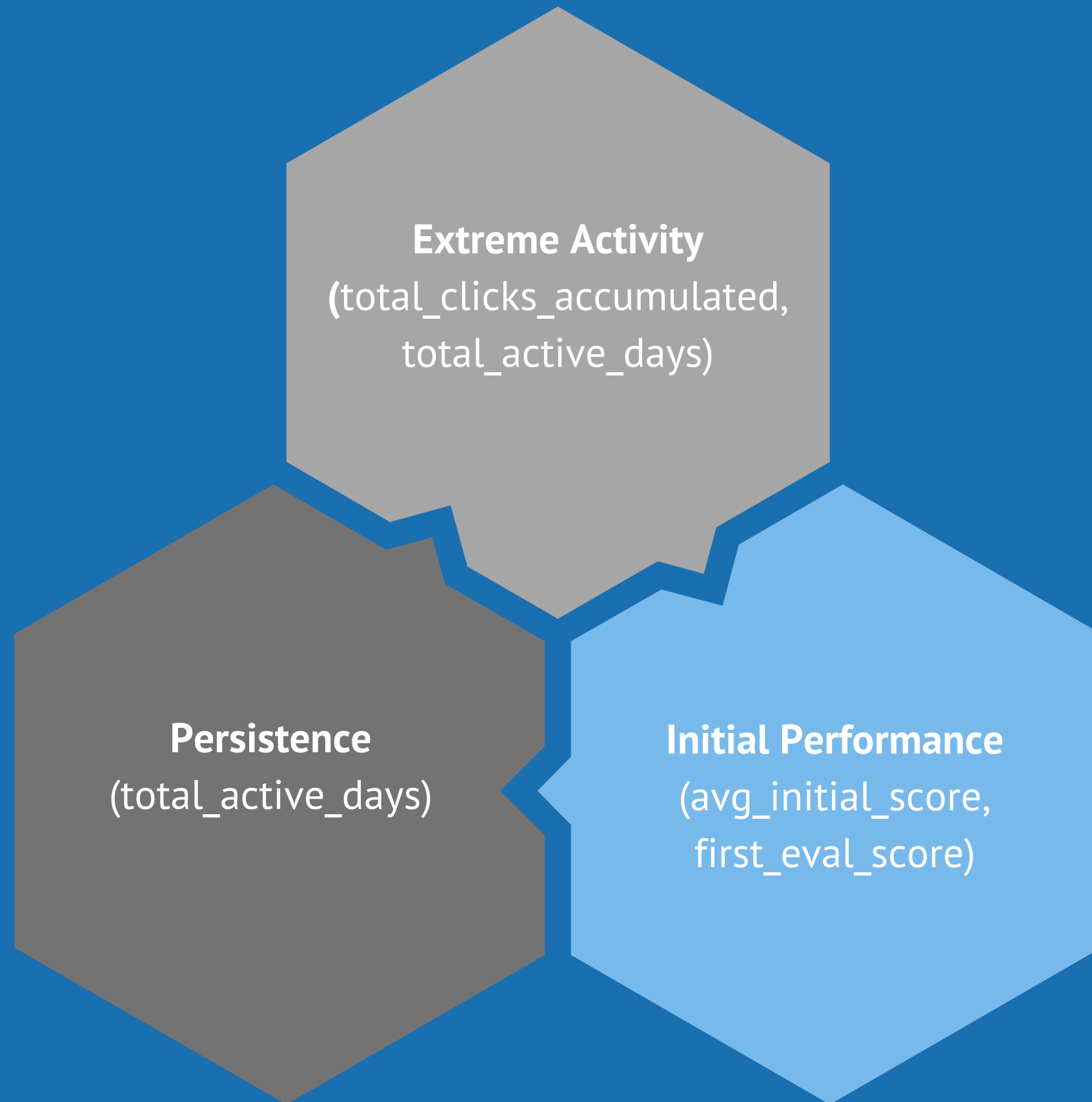
Inactive students and those who drop out early. The highest risk group.)

CLUSTER 3 (MODERATE RISK)

Intermediate scores, moderate clicks and active days

Students with a history of previous attempts who need intensive support.

Key Risk and Success Factors



Cluster 1
vs.
Cluster 0

Very low activity is the strongest predictor of failure (C1). Very high activity is the strongest predictor of success (C0).

Cluster 0 vs.
Cluster 2

For active students (C0, C2, C3), the score is crucial. A high initial score (C0) ensures success. A low initial score (C2) predicts failure despite an attempt at the activity.

Cluster 1 vs.
Cluster 2

Low persistence differentiates risk: C1 (2 days) has extreme risk, while C2 (15 days) and C3 (39 days) show an attempt at continuity that slightly reduces the risk, but it remains high if performance is low.

Descriptive Analysis of Student Clusters



CORRELATION BETWEEN ACTIVITY AND PERFORMANCE

Regression analysis showed a positive and significant correlation between the level of activity in the EVA and the probability of success.

LOW ACTIVITY (C1)

Very low activity, characterized by fewer than 100 accumulated clicks and a persistence of only 2 days, was the factor most strongly correlated with academic failure

HIGH ACTIVITY (C0)

Sustained, high activity was the most robust predictor of success

IMPORTANCE OF INITIAL PERFORMANCE

A critical interaction was observed between Activity and Initial Performance (first_eval_score). In students who did show activity (C0, C2, C3), the initial score was a determining factor in their performance.

Intervention Recommendations

Priority assignment of tutors to high demographic risk groups.

**ACTION 3
(DEMOGRAPHICS)**

ACTION 2 (BEHAVIOR)

Encourage VLE activity in the first few weeks by setting a minimum click target.

ACTION 1 (PERFORMANCE)

Strategies based on the initial score threshold.

Summary of Key Findings

- Extreme Risk of Disconnection (C1): persistence in the first few weeks is the strongest predictor, indicating a greater probability of failure. This group requires urgent reactivation intervention.
- The Persistence Paradox (C2): despite demonstrates moderate and persistent effort of activity, but fails due to a low initial performance. The initial score is the dominant risk factor for this motivated segment.
- Early Intervention Thresholds: two critical issues for immediate intervention:
 - 1) The first_eval_score (initial score)
 - 2) The total_clicks_acum and total_active_days in weeks mid-term weeksInterventions should be different: **academic** for low score and **engagement** for low activity.

Thanks for learning together



Juan Pablo Romagnoli

Ed Tech & Learning Analytics | Data
Analyst (SQL - Python - Power BI) | Ba...



*“Learning by
understanding focuses
more on the process of
learning something than
learning about something
(knowing how to do it)”.*

David Perkins